

MAE 140 Winter 12 Homework #6

TR & T, 9.1, 9.3, 9.8, 9.10, 9.11, 9.16, 9.20, 9.25, 9.26, 9.27

Due: 02/28/2012

9-1 Find the Laplace transform of $f(t) = 10[e^{-50t} - 2e^{-100t}]u(t)$. Locate the poles and zeros of $F(s)$.

9-3 Find the Laplace transform of $f(t) = 25[1 - 3\cos(500t)]u(t)$. Locate the poles and zeros of $F(s)$.

9-8 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.

(a) $f_1(t) = 4\delta(t) + [20e^{-20t} + 40e^{-40t}]u(t)$

(b) $f_2(t) = [15 + 15\cos 500t]u(t)$

9-10 Find the Laplace transforms of the following waveforms.

(a) $f_1(t) = 3\delta(t - 2)$

(b) $f_2(t) = 10e^{-50(t-1)}u(t - 1)$

(c) $f_3(t) = 10e^{-50(t-2)}u(t - 2)$

9-11 Use MATLAB to find the Laplace transform of the following waveform.

$$f(t) = [10 + 2e^{-10t}]u(t) + [5\cos 100(t - 0.05)]u(t - 0.05)$$

9-16 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{s + 100}{s(s + 50)}$

(b) $F_2(s) = \frac{s + 10}{s(s + 50)(s + 100)}$

9-20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{\alpha^2}{s^2(s + \alpha)}$

(b) $F_2(s) = \frac{\alpha^2}{s(s + \alpha)^2}$

9-25 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{4(s^2 + 16)}{s(s^2 + 8s + 32)}$

(b) $F_2(s) = \frac{2(s^2 + 30s + 800)}{s(s^2 + 50s + 400)}$

9-26 Find the inverse transforms of the following function and validate your answers using MATLAB:

(a) $F_1(s) = \frac{(s + 50)^2}{(s + 20)^2(s + 100)}$

(b) $F_2(s) = \frac{(s + 20)^2}{(s + 50)^2(s + 100)}$

9-27 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

(a) $f(t) = \delta(t) - 5e^{-20t}$

(b) $f(t) = \delta(t)$

(c) $f(t) = \delta(t) + 5e^{-20t}$